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Evaluation of SAFE-SEM Approach for Modeling Sonic Logging

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Abstract

The aim of this work is to analyze the performance of the semi-analytical finite element – spectral element method (SAFE-SEM) for computing dispersion curves of borehole modes for various formations. The modes of interest for sonic logging – Stoneley, flexural and quadrupole – are analyzed. The range of formations covers a range of slowness values and anisotropy (i.e., VTI, HTI, TTI). The dispersion curves are computed on meshes of different sizes and number of elements, and the calculations are carried out by both SAFE-SEM and traditional approaches. The results are compared to evaluate SAFE-SEM's performance.

SAFE-SEM is capable of using curvilinear elements. As a result, the domain can be approximated with a smaller number of elements. The possibility to use tool model in the form of the fluid column (wireline tools) or the pipe (LWD tools) has been confirmed as well. The method can handle both fast and slow formations. The performance is good and does not depend on the formation type – isotropic, VTI, HTI, or TTI. The results are of similar quality to those of the traditional SAFE approach, which makes it possible to recommend SAFE-SEM for use in sonic logging data interpretation algorithms.

Introduction

Sonic logging is an important geophysical exploration method founded on the propagation of acoustic waves through subsurface formations (Donald et al., 2013; Fang et al., 2015; Prioul et al., 2004; Prioul and Lebrat, 2004; Rider and Kennedy, 2011; Sinha and Kostek, 1996; Zharnikov and Nikitin, 2020). Processed acoustic data are crucial inputs for the computation of fundamental rock parameters like mechanical strength, porosity, in-situ stress levels, and fluid saturation. These calculated parameters play key role in forecasting wellbore stability—preventing collapse or unwanted fractures while drilling—and optimal planning for reservoir stimulation and subsurface stress field mapping. The mechanical properties of rock serve as one of the important features to differentiate the lithology. They are also excellent indicators of gas-bearing zones both on logs and seismic images. Fracture networks significantly affect mechanical properties of the formation and can be characterized by analyzing these properties. All this information is very important for reservoir characterization. Advances in sonic logging technologies have significantly improved formation evaluation accuracy to enable more precise reservoir characterization.

One of the techniques of processing and analyzing sonic logging data is the dispersion curve analysis. This is a method which begins with a collection of recorded waveforms in the time-depth domain and converts them to the frequency domain via Fourier analysis. The resulting dispersion diagram illustrates how phase velocity of guided acoustic waves varies with frequency in the borehole medium. To properly interpret these diagrams, forward modeling is necessary. Several methods can be used to compute synthetic dispersion curves, including: analytical solutions for isotropic and vertical transverse isotropic (VTI) media, matrix-based descriptions for cylindrically layered VTI environments, perturbation theory for small deviations from idealized conditions and direct 3D numerical modeling for complex geometries.

By correlating calculated dispersion curves with these theoretical models and minimizing discrepancies, geoscientists can make firmer inferences regarding formation properties. All standard techniques, however, are contingent on a perfect circular borehole and minimum anisotropy, assumptions frequently violated by heterogeneous or highly anisotropic formations. While 3D modeling is more accurate, its computational cost typically makes routine runs unacceptable. Consequently, greater emphasis is given to alternative techniques as accurate as possible but efficient.

One candidate of particular interest is the Semi-Analytical Finite Element (SAFE) approach, which has also been advocated as a forward modeling technique for dispersion curves analysis (Ellefsen et al. (1991); Zharnikov and Syresin (2015)). SAFE comes in most handy in those cases where analytical solutions are non-existent. Recently, the technique has been perfected through the incorporation of spectral elements (Zharnikov et al., 2022), an alternative form of finite element analysis that uses high-order nodal basis functions (Komatitsch and Vilotte, 1998; Charara and Sabitov, 2020). The SEM gives two major advantages: discrete orthogonality, which minimizes computations by creating block-diagonal mass matrices, and high-order approximation precision, such that precision is gained in modeling using fewer elements. Additionally, SEM supports curvilinear meshes, enabling more efficient representation of borehole geometry while maintaining computational tractability.

To ensure the effectiveness of SEM-based SAFE modeling, a detailed analysis of its sensitivity to various parameters is required. The most crucial of these include the choice of linear vs. curvilinear elements, polynomial approximation order, and the number of elements used for discretization. Furthermore, an assessment of the robustness of the method against diverse real-world formation properties is absolutely essential to verify practical applicability. Here, we address these questions in this study by applying SAFE-SEM to a list of benchmarking formations documented in the literature. Through the computation of dispersion curves for these examples, we determine the merits and shortcomings of the approach with relevance to its field-scale sonic log interpretation optimization.

Theory and methodology

We use semi-analytical finite element method (Ellefsen et al., 1991; Zharnikov and Syresin, 2015). It applies Fourier transform to the Helmholtz equation and discretizes it in 2d (x, y) plane for given wavevector k. The displacement field assumes the form:

$$\boldsymbol{u}(x, y, k, \omega) = \boldsymbol{N}(x, y)\boldsymbol{u}(k, \omega)e^{i(kz-\omega t)} \tag{1}$$

Here, $\boldsymbol{u}(k,\omega)$ is the displacement and $\boldsymbol{N}(x,y)$ is the basis function. Putting it into weak formulation of the Helmholtz equation, one arrives at the following equation:

$$\left(\mathbf{K}_1 + ik\mathbf{K}_2 + k^2\mathbf{K}_3 - \omega^2\mathbf{M} + i\omega\boldsymbol{P}\right)\mathbf{U} = \mathbf{0} \tag{2}$$

which is the generalized eigenvalue problem. The matrix coefficients entering into Eq (2) are computed as the integrals of various products of basis functions, their derivatives and physical properties like density and elastic moduli. For example:

$$\mathbf{K}_1^e = \int\limits_V^{(e)} \mathbf{B}_1^* \mathbf{C} \mathbf{B}_1 dV \qquad \mathbf{M} = \int\limits_V^{(e)} \mathbf{N}^* \rho \mathbf{N} dV \tag{3}$$

Any elements and basis functions can be used to implement SAFE approach. Here, we utilize curvilinear spectral elements. It uses Lagrange polynomials as the basis functions. They are equal to 1 at the specific node and zero in others (see Figure 1). The nodes of Gauss-Lobatto-Lagrange (GLL) quadrature are used to define the positions of the element's nodes. As a result, the computation of the finite element matrices is greatly simplified, and the mesh size can be significantly reduced. For the computations on unstructured grids, the transformation from the reference cell into the physical space is non-linear. The following map is used:

$$\begin{cases} x = a_x + b_x\xi + c_x\eta + d_x\xi\eta + e_x\xi^2 + f_x\eta^2 + g_x\xi^2\eta + h_x\eta^2\xi \\ y = a_y + b_y\xi + c_y\eta + d_y\xi\eta + e_y\xi^2 + f_y\eta^2 + g_y\xi^2\eta + h_y\eta^2\xi \end{cases} \tag{4}$$

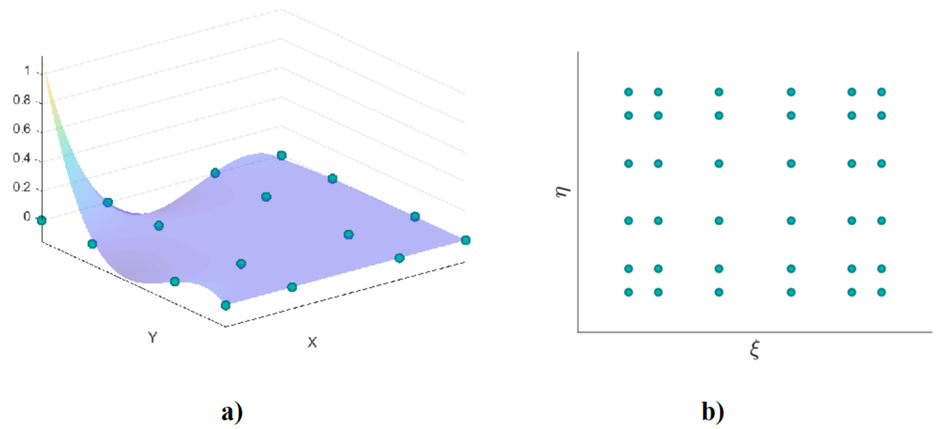


Figure 1—Example of basis function (a) and nodes (b).

Using it to compute the matrix coefficients of Eq (2), the following expressions are obtained:

$$M_{ij} = \int N_i(x, y) N_j(x, y) dx dy = \int N_i(\xi, \eta) N_j(\xi, \eta) J d\xi d\eta \tag{5}$$

Here, J is the determinant of the Jacobi matrix

$$J = \begin{vmatrix} \partial x / \partial \xi & \partial x / \partial \eta \\ \partial y / \partial \xi & \partial y / \partial \eta \end{vmatrix} \tag{6}$$

The dispersion curves can be obtained by solving Eq. (2) for a set of the wavevectors. Many algorithms exist, which can be used to solve Eq. (2).

To apply SAFE methodology to sonic logging, one has to construct 2D mesh to approximate borehole cross section. It can be derived, for example, from caliper measurements. Often, circular cross section is assumed. If more detailed information, like multi-arm calipers, is available, the form can be made closer to the reality. An example of the mesh is shown in Figure 2. After setting up the mesh and the parameters, the dispersion curves can be computed and compared with the results of the measurements. To use the method to interpret field data, it has to be sufficiently fast accurate. In the next section, we investigate the performance of the method for different values of the parameters and test it on the models of the formations, for which the values of the elastic moduli were reported in the literature.

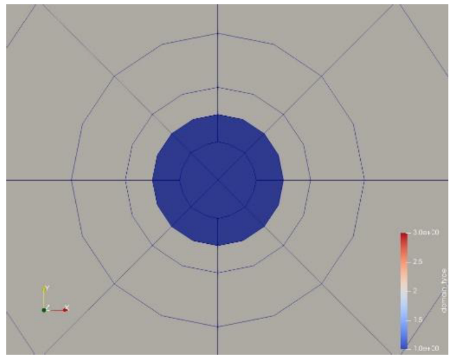


Figure 2—Computational mesh.

Method's evaluation and discussion

The computational efficiency of the SAFE-SEM methodology was evaluated systematically in two ways. First, we investigated the dependence of method's performance and accuracy on such parameters as the number of the nodes in the mesh, degree of approximating polynomial and others. Second, several formations were considered and the dispersion curves computed. As the reference, we took the dispersion curves computed by SAFE method, which was thoroughly tested before.

Numerical results for a range of formations are presented below. The elastic parameters are listed in Table 1. These are Cotton Valley shale, Bakken shale, Barnett shale (Thomsen, 1986; Norris and Sinha, 1995; Sinha et al., 2016). Borehole was assumed to be filled with water with the bulk modulus $K = 2.25\ GPa$ and the density $\rho = 1\ g/cc$. The borehole diameter was assumed to be 8 inches. The dispersion curves were computed in the range of frequencies from 1 to 15 kHz with 1 kHz step, matching the standard operating window for sonic logging applications. To cope with the finite size of the numerical model, absorbing boundary conditions were used. In addition, the outer radius of the model was adjusted with the frequency to ensure correct computation of the dispersion curves in the whole frequency range. The shales were modeled as tilted transverse isotropy (TTI) medium with the vertical transverse isotropy (VTI) symmetry axis tilted at an angle α to the borehole axis. The borehole inclination of 0, 30, 45 and 60 degrees was modeled. Horizontal transverse isotropy (HTI) case corresponding to $\alpha = 90$ was modeled as well.

Table 1—Formation properties.

	ρ, kg/m^3	c_{11}, GPa	c_{13}, GPa	c_{33}, GPa	c_{44}, GPa	c_{66}, GPa
Cotton Valley shale	2640	74.73	25.29	58.84	22.05	29.99
Bakken shale	2230	40.9	8.5	26.9	10.5	15.3
Barnett shale	2500	54.872	11.269	36.1	14.4	19.584
Water	1000	-	-	-	-	-

The analysis of computation time taken to calculate dispersion curves for various polynomial orders p was conducted using curvilinear spectral elements with constant azimuthal discretization n_{az} and varied the radial cell number n_r. The total number of degrees of freedom was approximately the same. The tests demonstrated an exponential growth in computation time with polynomial order. The convergence analysis was done for two test cases with well deviation angles of 0 and 45 degrees. The grid refinement analysis used mesh doublings. It showed the efficiency of SAFE-SEM compared to the conventional SAFE approach.

Comprehensive accuracy assessment was done as follows. First, it investigated two important parameters: polynomial order p and n_{az}. This two-parameter study revealed several important trends as follows. For p ranging from 2 to 9, solution accuracy was within 2 μs/ft for frequencies greater than 2 kHz (Figure 3).

Counterintuitively, higher-order elements gave poorer performance for Stoneley mode at low frequencies (1-2 kHz). Plausible explanation is the poor radial resolution (only 1-2 cells) at high p. The optimal balance between accuracy and efficiency appears to be at p = 2. Analysis of the accuracy behavior with varying n_{az} indicated that a minimum of 8 azimuthal cells $n_{az} \geq 8$ is required for acceptable accuracy. Coarser discretizations results in large errors up to 5 μs/ft for Stoneley modes and significant deviations near flexural mode inflection points (Figure 4). Increasing n_{az} beyond 8 gave no visible benefits, only computational cost.

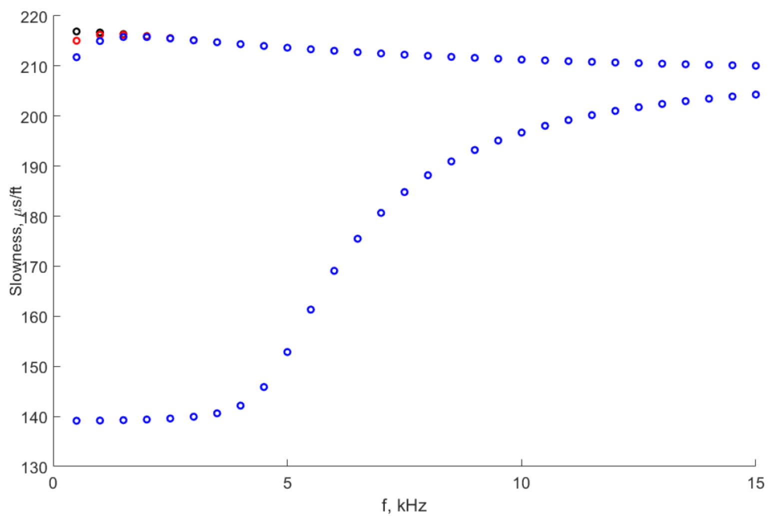


Figure 3—Dispersion curves for VTI Bakken shale for p=2 (black), p=5 (red) and p=9 (blue).

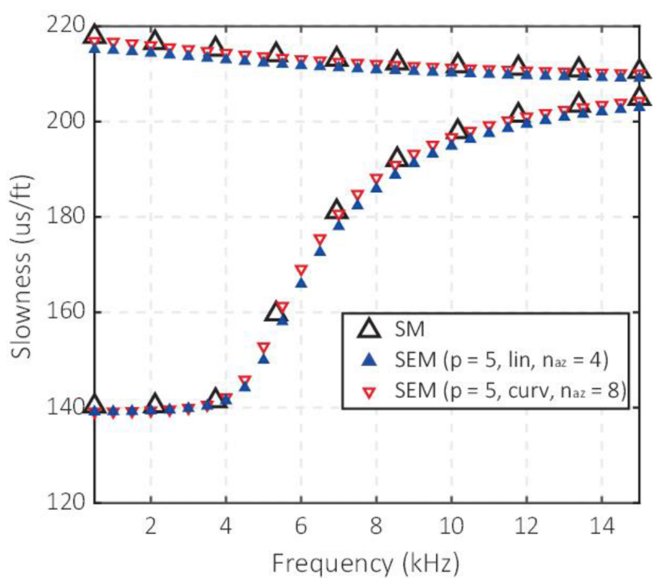


Figure 4—Dispersion curves for VTI Bakken shale for n_{az} = 4(black) and n_{az} = 8 (red).

The analysis of linear versus curvilinear element type showed that the former can be used for adequate Stoneley mode characterization, for which the errors do not exceed 1 μs/ft (Figure 5). But linear elements lack for dipole modes giving the errors in mid-frequency up to 3 μs/ft. This is most likely due to insufficient geometric fidelity. The situation is corrected by curvilinear elements, which maintain accuracy even for coarse grids. It comes at marginal computational overhead compared to linear elements.

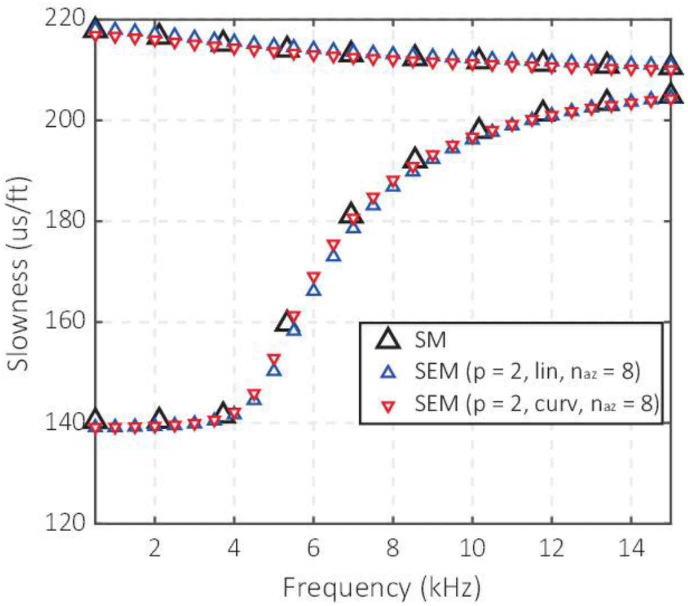
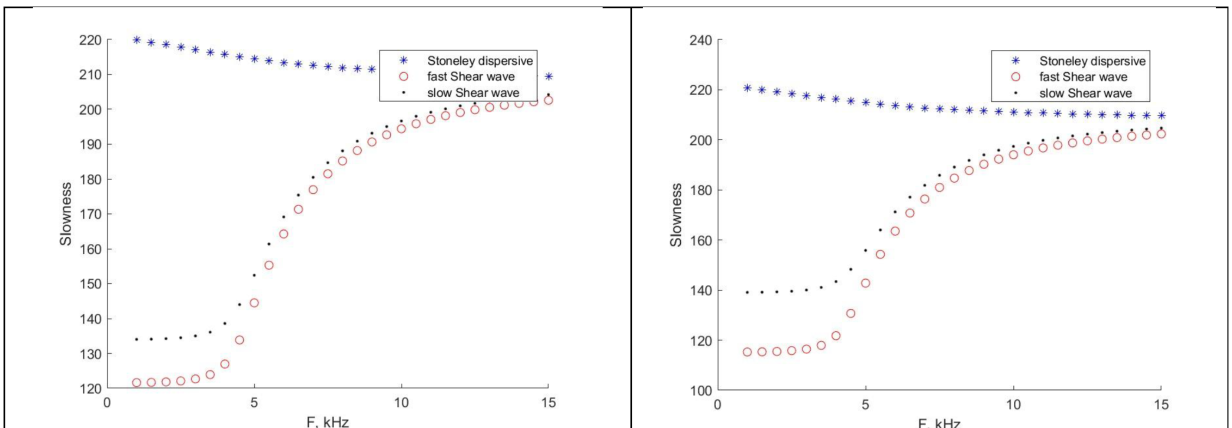


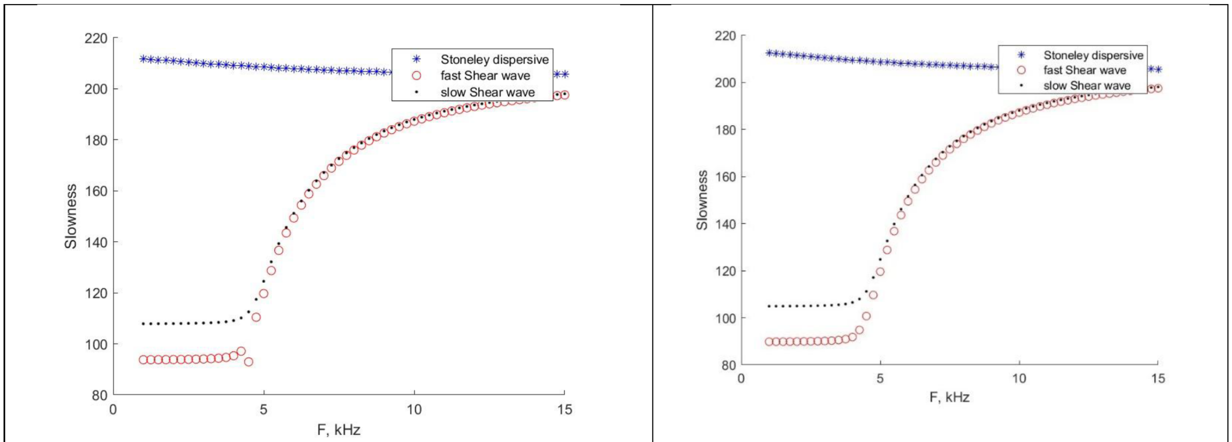
Figure 5—Dispersion curves for VTI Bakken shale for n_{az} = 8 computed using linear (blue) and curvilinear (red) elements.

The second part of method's evaluation was to validate it for different formations. The dispersion curves were computed for the formations listed in Table 1. Considering the accuracy analysis above, the computations were done using curvilinear elements with p = 2 and n_{az} = 8. The results are presented in Figures 6-8. In all cases, good agreement with the reference dispersion curves was observed. The discrepancy did not exceed 1 µs/ft.



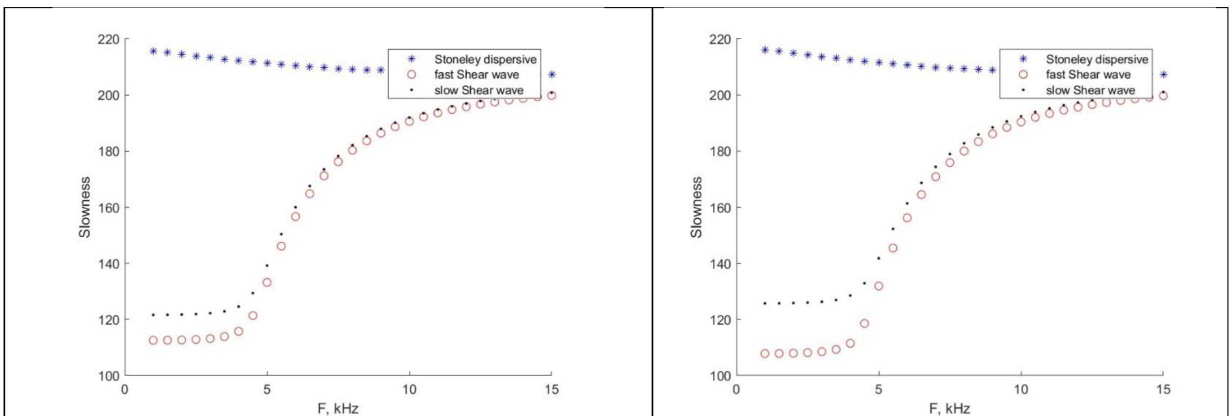
(a) (b)

Figure 6—Dispersion curves for Bakken shale for wells: a) deviated (60°); b) horizontal.



(a) (b)

Figure 7—Dispersion curves for Cotton Valley shale for wells: a) deviated (60°); b) horizontal.



(a) (b)

Figure 8—Dispersion curves for Barnett shale for wells: a) deviated (60°); b) horizontal.

Conclusions

This work gave a detailed examination of the performance characteristics of the SAFE-SEM method, its testing in various model scenarios and parameter sets. A careful test was conducted to examine the sensitivity of the technique to some of the principal parameters discussed below. It examined how the changing polynomial orders in the spectral element configuration affect the solution accuracy. High-order polynomials are likely to provide superior wavefield feature resolution but are computationally expensive. Particular attention was given to the mesh topology – number and arrangement of the elements used to discretize borehole geometry radially and azimuthally. Linear vs. curvilinear element performance comparison revealed important accuracy vs. computational expense trade-offs, particularly when simulating complex borehole geometries. The overall benchmarking demonstrated high consistency between SAFE-SEM solutions and reference solutions for a wide set of parameters. The method correctly simulated important borehole modes (Stoneley, flexural, and screw mode) and higher-order guided waves. Notably, the method performed reliably when simulating challenging transversely isotropic media with tilted symmetry axes (TTI media), including Bakken Shale, Barnett Shale and Cotton Valley Shale.

A key finding was that the method could be used to generate high-fidelity dispersion curves using very coarse discretization – even as coarse as nine elements for approximating the borehole. This is a significant advantage over more conventional methods that would normally require much finer meshes in order to achieve similar levels of accuracy. The method's robustness was also demonstrated through validation tests on non-circular borehole geometries (Zharnikov and Syresin, 2015), demonstrating capacity to deal with real wellbore complexities often achieved in field operations. It is therefore best applied for the interpretation of sonic logging data, particularly when correct modeling of dispersion curves in complicated anisotropic formations or curved boreholes is essential.

The SAFE-SEM approach is a viable replacement for standard SAFE formulations in forward sonic logging modeling. It generates dispersion curves of the same quality as the standard SAFE method but with coarser grids and diminishing computational demands. This upgraded setup has the ideal accuracy-to-efficiency ratio and is very efficient at real-time interpretation of logs in advanced anisotropic formations.

The SAFE-SEM technique offers a number of distinctive advantages for sonic logging interpretation as outlined below. The method is good at modeling wave propagation in complex anisotropic media to generate realistic dispersion curves required for effective formation evaluation. Its ability to handle non-circular cross-sections of the borehole renders it most valuable in real logging practice where shape irregularities of the borehole are common. Its balance of accuracy and efficiency in computation makes SAFE-SEM an ideal candidate for real-time logging data interpretation systems.

As compared to typical SAFE formulations, the SAFE-SEM approach computes comparable dispersion curves to conventional SAFE implementations but achieves these results with coarse computational meshes, lowers memory requirements and processing time. This performance profile is due to the method's optimum spectral element formulation, intelligent mesh design and optimized numerical solution. For applied field operations in practice, the method has distinct benefits in challenging reservoirs, where high-order anisotropy significantly affects acoustic measurements. Furthermore, it can be used in horizontal and deviated wells, where accurate wave propagation modeling is required to be modeled and for operations, where real-time interpretation of sonic data is critical. The trade-off between computation speed and physical correctness in this method makes it ideally suited to logging-while-drilling (LWD) operations, advanced formation evaluation and integrated interpretation procedures.

The extensive examination confirms that SAFE-SEM represents a significant improvement in borehole acoustic modeling. The ability to reconcile computational cost with physical accuracy, particularly in challenging anisotropic conditions, makes it a versatile tool for present-day sonic logging interpretation. The insensitivity of the approach to parameters and to realistic borehole geometries solidifies it as a desirable choice for both research and production environments within the petroleum industry.

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